QP CODE: 25900156

SERIES :A

A

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025
SECOND SEMESTER
Multi-Disciplinary Courses (MDC) - MG2MDCPHY100 - OBSERVATIONAL ASTRONOMY
(2024 ADMISSION ONWARDS)

Duration: 1 Hours **Maximum Marks: 35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overalloutcome attainability.

[Learning Domain] [CO No(s)]

Part A
Multiple Choice Questions
Answer all questions
Each question carries **1 mark**

1.	What is the approximate age of the Universe in years according to the Big Bang theory? *Ans:***C** **13.8 billion**	[K] / [CO1]
2.	Which of the following is a unit used to express the size of our solarsystem? *Ans:***B** **Astronomical Unit (AU)**	[K] / [CO1]
3.	Which planet is closest to sun in heliocentric model? *Ans:***C** **Mercury**	[K] / [CO1]
4.	What is the duration of one Solar day? *Ans:***B** **24 hours**	[K] / [CO4]
5.	The apparent path of the Sun against the background stars is known as: *Ans:***B** **Ecliptic**	[K] / [CO4]
6.	Which part of a refractive telescope magnifies the image? *Ans:***C** **Eyepiece Lens**	[K] /[CO2]
7.	Where is the location of James web Telescope? *Ans:***C** **Approximately 15 lakh km from earth**	[K] / [CO2]
8.	Magnifying Power of a telescope is defined as the ratio of: *Ans:***A** **Focal length of Objective Lens to Focal length of Eyepiece Lens**	[K] / [CO2]

9.	Eventually what does a white dwarf become? *Ans:***D** **A black dwarf**	[K] / [CO3]
10.	A collapsed core of a massive star after supernova is called as: *Ans:***D** **Neutron Star**	[K] / [CO3]
11.	What is the most famous nebula visible to the naked eye, known for star formation? *Ans:***A** **Orion Nebula**	[K] / [CO3]
12.	Who classified galaxies based on their shapes? *Ans:***B** **Edwin Hubble**	[K] / [CO4]
13.	When does a full lunar eclipse occur? *Ans:***D** **When earth passes directly in front of full moon**	[K] / [CO5]
14.	Which is the nearest planet to sun? *Ans:***A** **Mercury**	[K] / [CO5]
15.	Where is the location of main Asteroid belt in the Solar System? *Ans:***B** **Between Mars and Jupiter**	[K] / [CO5]

[1x15 = 15]

Part B

Multiple Choice Questions

Answer all questions

Each question carries **2 marks**

16.	Which method was used by Galileo Galilei to test hypotheses? *Ans***: C** **Systematic observation and experimentation**	[K] / [CO1]
17.	What does the term "deferent" refer to in the Ptolemaic model? *Ans:***C** **The larger circle along which the center of the epicycle moves**	[K] / [CO1]
18.	What is the brightest star in the constellation of Orion? *Ans:***B** **Rigel**	[K] / [CO4]
19.	Which electromagnetic radiation is used to detect quasars and pulsars? *Ans:***B** **Radio Waves**	[K] / [CO2]
20.	How often does Hubble Space Telescope orbits earth? *Ans:***C** **Every 90 mins**	[K] / [CO2]
21.	Which colour is most associated with the hottest stars? *Ans:***B** **Blue**	[K] / [CO3]

22.	Which of the following best describes the core condition of a red giantstar? *Ans:***B** **The core is composed of helium and contracts while the outer layers expand**	[U] / [CO3]
23.	At what approximate speed does the Sun revolves around the centre ofMilky way galaxy? *Ans:***B** **250 km/s**	[K] / [CO4]
24.	Why doesn't solar eclipse occur every month? *Ans:***C** **The Moon's orbit is tilted**	[K] / [CO5]
25.	Which of the statement about sun is wrong? *Ans:***D** **Main component of sun is helium**	[U] / [CO5]

[2x10 = 20]